

6: EXPERIMENT-16 16 BIT SUBTRACTION

AIM: TO write an assembly language program to implement 16 Bit subtraction using 8085

ALGORITHM:

1. Start the program by loading a register pair with address of 1st number.
2. copy the data to another register pair
3. Load the second number to the first register pair
4. sub the two register pair contents
5. check for carry
6. store the value of sum and carry.
7. END.

PROGRAM:

MEM MONICS	EXPLANATION.
LHLD 2050	Load the 16 Bit data from memory locations 2050H and 2051H.

X C H G

LHLD 2052

MVIC, 00

MOVA, E

SUBL

STA 2054

MOV, A, D

SUB H

HLT

Exchange the contents of HL and DE register pairs.

Load the 16 Bit data from the memory.

move the immediate value 00H into

move the counter to register into A, E.

Subtract the content of register L from.

store the result.

move the content of register to H format.

subtract the content of register H.

Halt.

INPUT:

Address	Data
2054	2
2052	3

out put:

Address.	Data
2054	1
2055	1

RESULT:

Thus the program was executed successfully using 8085 processor simulator

File Reset Assembler Debug Help



Registers		Flag
A	02	S 0
BC	00 00	Z 0
DE	02 00	AC 0
HL	00 03	P 0
PSW	00 00	C 0
PC	42 14	
SP	FF FF	
Int-Reg	00	

Decimal - Hex Conversion

Decimal	Hex
0	0

To Hex To Dec

I/O Ports

0 - + 00

Update Port Value

Memory

2055 - + 1

Update Memory

Load me at

```
1 LHL D 2050
2 XCHG
3 LHL D 2052
4 MVI C, 00
5 MOV A, E
6 SUB L
7 STA 2054
8 MOV A, D
9 SUB H
10 STA 2055
11 HLT
```

Data Stack KeyPad Memory I/O Ports

Start 2050 OK

Address (Hex)	Address	Data
0802	2050	0
0803	2051	2
0804	2052	3
0805	2053	0
0806	2054	1
0807	2055	1
0808	2056	0
0809	2057	0
080A	2058	0
080B	2059	0
080C	2060	0
080D	2061	0
080E	2062	0
080F	2063	0

Line No Assembler Message

0 Program assembled successfully